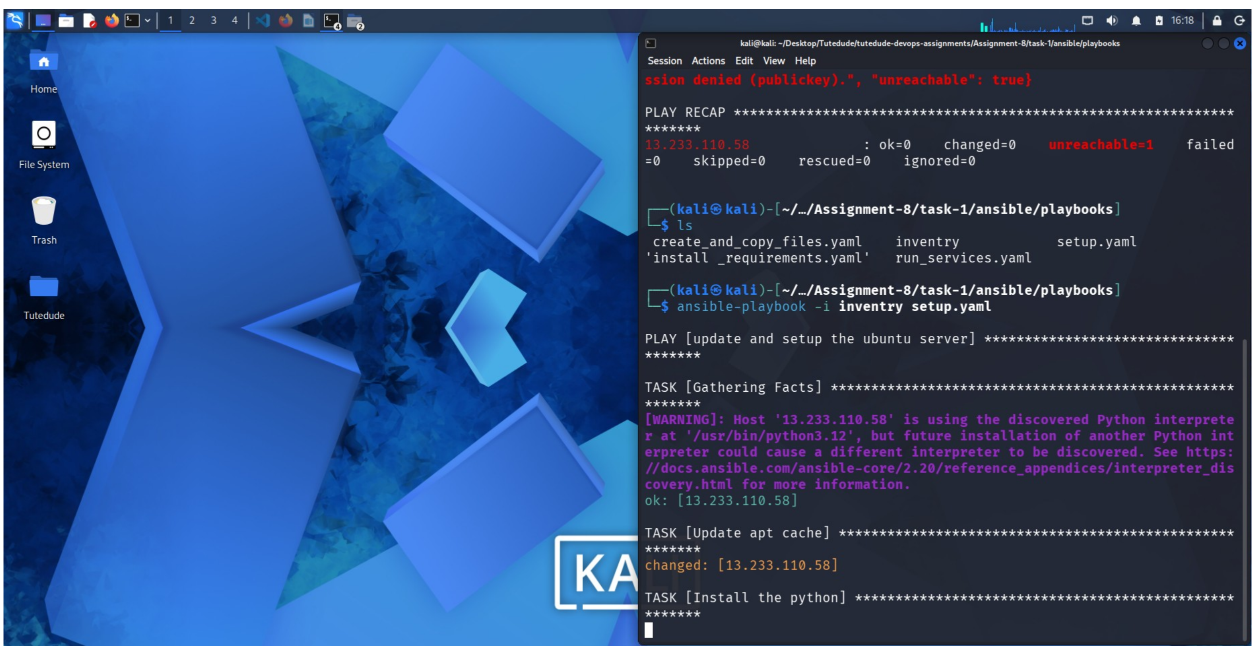
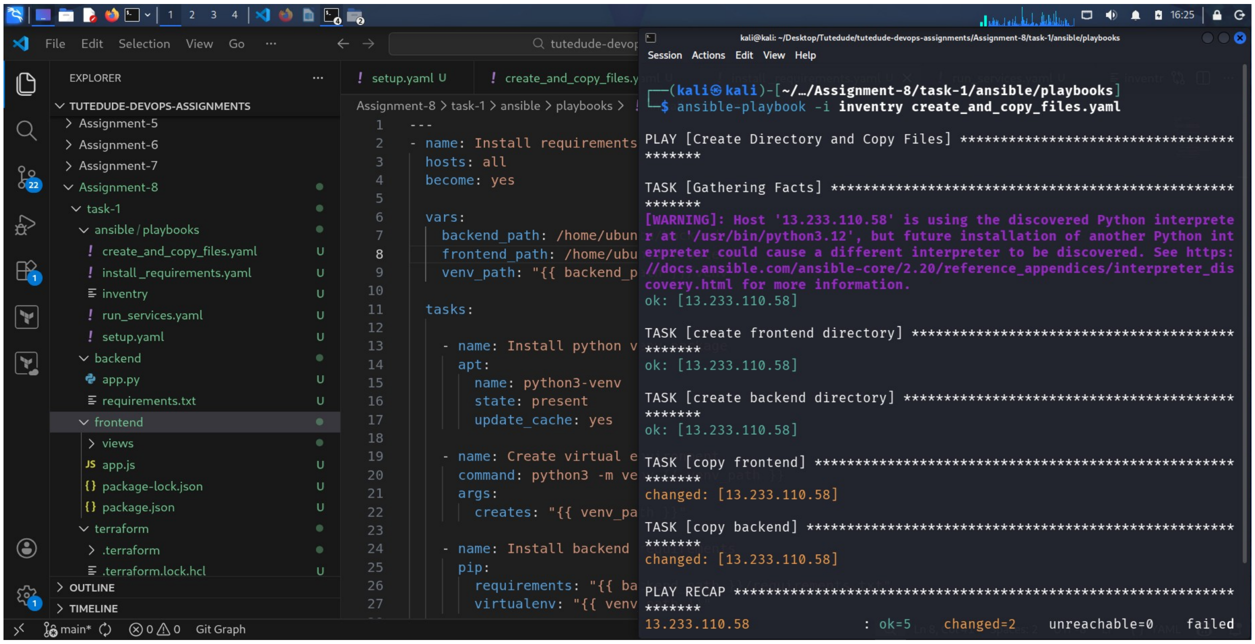
You are required to deploy a Flask backend and an Express frontend application using **AWS** and **Terraform**. Follow the steps below to deploy the applications in different configurations.

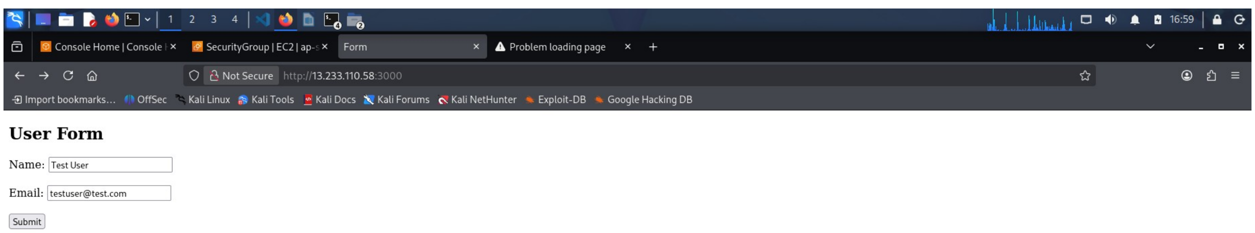
Part 1: Deploy Both Flask and Express on a Single EC2 Instance

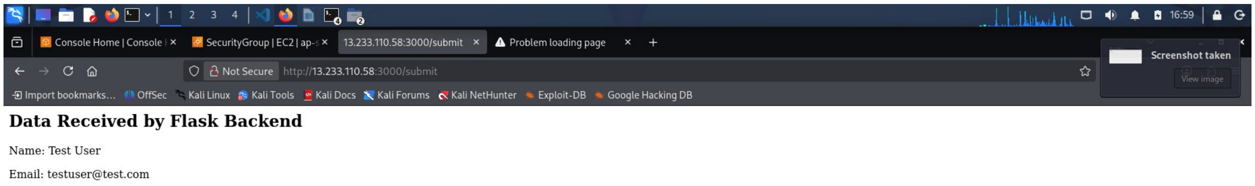
1. **Objective**: Deploy both the Flask backend and the Express frontend on a **single EC2 instance** using Terraform.
   - Provision the EC2 instance with Terraform.
   - Configure the instance with:
     - A user data script or a configuration management tool (e.g., Ansible or Cloud-Init) to install dependencies (Python, Node.js).
     - Scripts to start the Flask backend and the Express frontend.
   - Ensure both applications are running on different ports (e.g., Flask on port 5000 and Express on port 3000).
2. **Expected Deliverables**:
   - Terraform configuration files (main.tf, variables.tf, etc.).
   - A working EC2 instance with Flask and Express running and accessible via the instance's public IP.

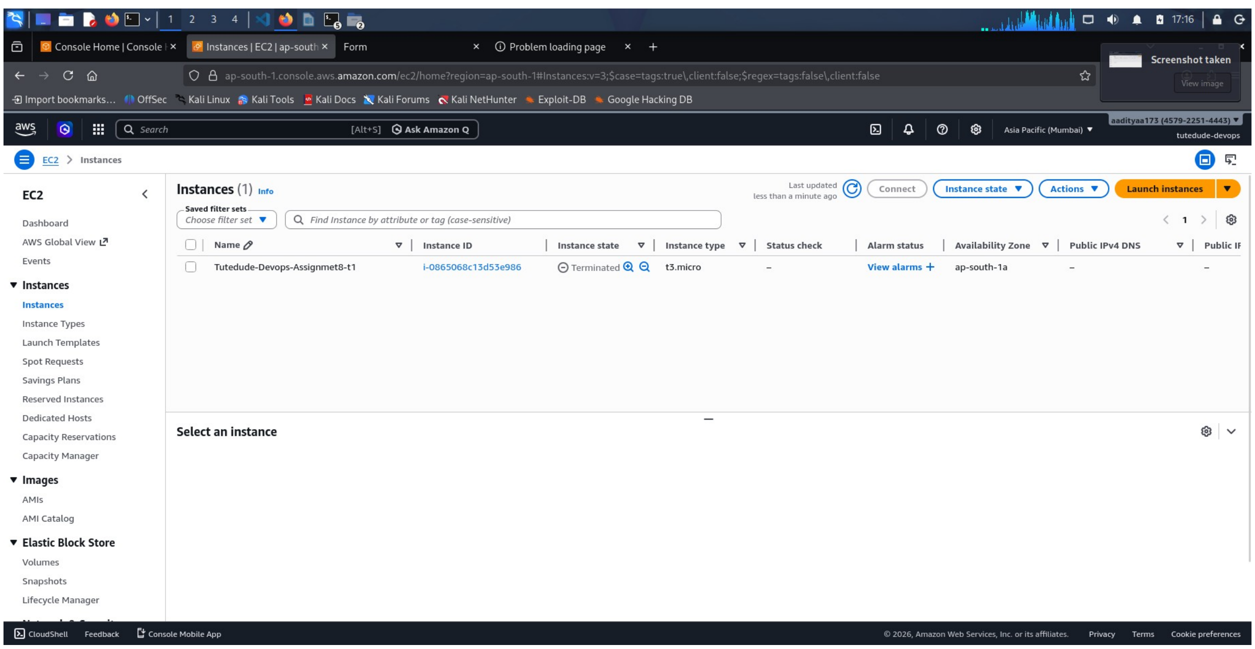



File Edit Selection View Go
EXPLORER
TUTEDUDE-DEVOPS-ASSIGNMENTS
Assignment-5
Assignment-6
Assignment-7
Assignment-8
task-1
ansible / playbooks
create_and_copy_files.yaml
install_requirements.yaml
inventry
run_services.yaml
setup.yaml
backend
app.py
requirements.txt
frontend
views
app.js
package-lock.json
package.json
terraform
.terraform
.terraform.lock.hcl
OUTLINE
TIMELINE
main*
Git Graph
setup.yaml U
create_and_copy_files.y
Assignment-8 > task-1 > ansible > playbooks >

- name: Install requirements
hosts: all
become: yes
vars:
backend_path: /home/ubun
frontend_path: /home/ubu
venv_path: "{{ backend_p
tasks:
- name: Install python v
apt:
name: python3-venv
state: present
update_cache: yes
- name: Create virtual e
command: python3 -m ve
args:
creates: "{{ venv_pa
- name: Install backend
pip:
requirements: "{{ ba
virtualenv: "{{ venv
kali@kali: ~/Desktop/Tutedude/tutedude-devops-assignments/Assignment-8/task-1/ansible/playbooks
Session Actions Edit View Help
(kali㉿kali)-[~/…/Assignment-8/task-1/ansible/playbooks]
$ ansible-playbook -i inventry create_and_copy_files.yaml
PLAY [Create Directory and Copy Files] ***
TASK [Gathering Facts] ***
[WARNING]: Host '13.233.110.58' is using the discovered Python interpreter at '/usr/bin/python3.12', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [13.233.110.58]
TASK [create frontend directory] ***
ok: [13.233.110.58]
TASK [create backend directory] **
ok: [13.233.110.58]
TASK [copy frontend] ***
changed: [13.233.110.58]
TASK [copy backend] **
changed: [13.233.110.58]
PLAY RECAP ***
13.233.110.58 : ok=5 changed=2 unreachable=0 failed


Console Home | Console
SecurityGroup | EC2 | ap-
Form
Problem loading page
Not Secure
http://13.233.110.58:3000
Import bookmarks… OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
User Form
Name: Test User
Email: testuser@test.com
Submit


Console Home | Console
SecurityGroup | EC2 | ap-
13.233.110.58:3000/submit
Problem loading page
Screenshot taken
View image
Not Secure
http://13.233.110.58:3000/submit
Import bookmarks... OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
Data Received by Flask Backend
Name: Test User
Email: testuser@test.com


Console Home | Console
Instances | EC2 | ap-south
Form
Problem loading page
Screenshot taken
View image
ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#Instances:v=3;$case=tags:true\,client:false;$regex=tags:false\,client:false
Search
[Alt+S]
Ask Amazon Q
Asia Pacific (Mumbai)
tutedude-devops
EC2
Instances
Instances (1)
Info
Last updated
less than a minute ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Find Instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4 DNS
Tutedude-Devops-Assignmet8-t1
i-0865068c13d53e986
Terminated
t3.micro
View alarms
ap-south-1a
Dashboard
AWS Global View
Events
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Select an instance
CloudShell
Feedback
Console Mobile App
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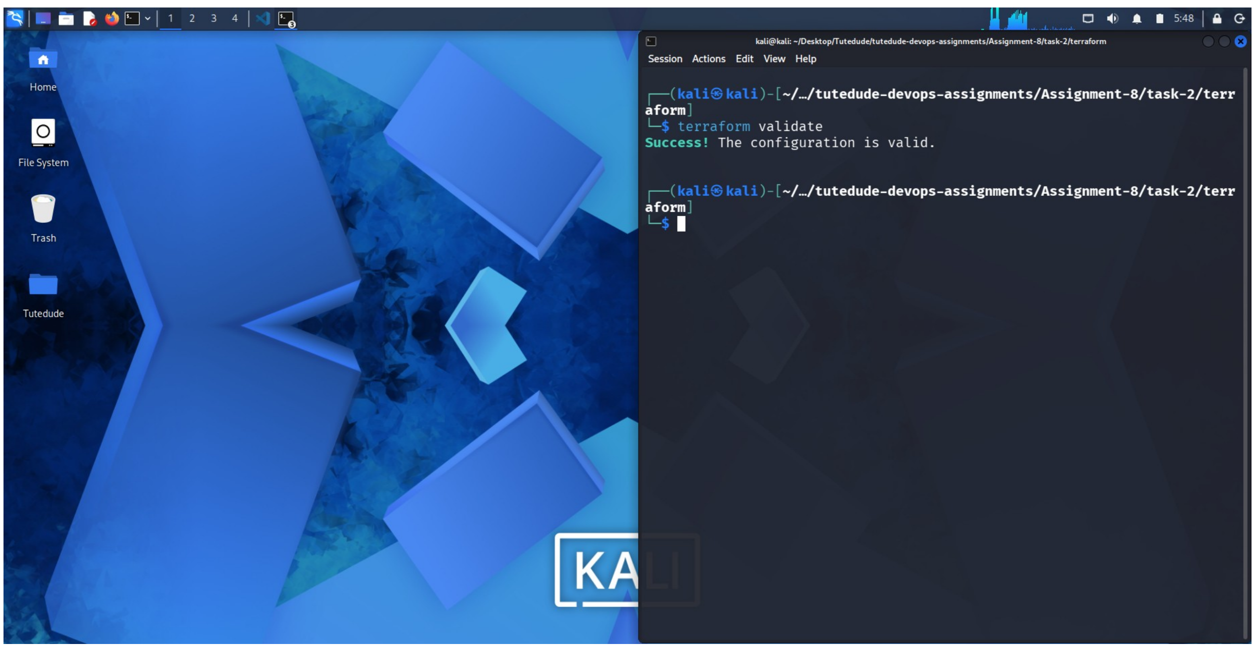
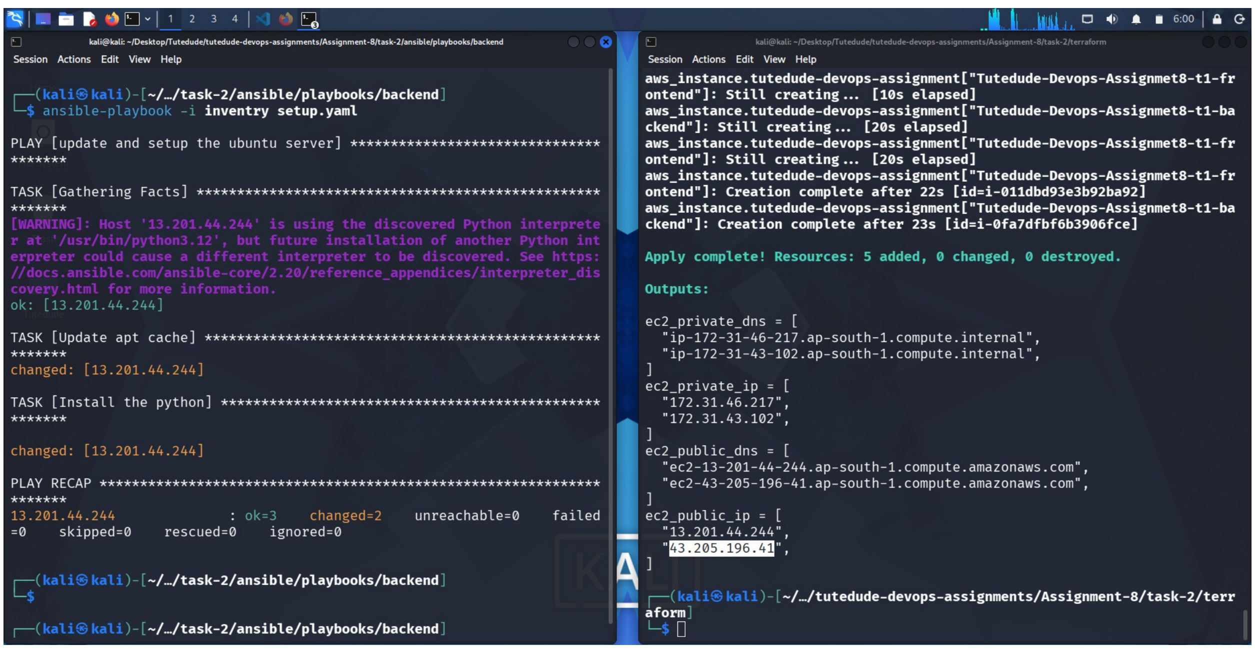
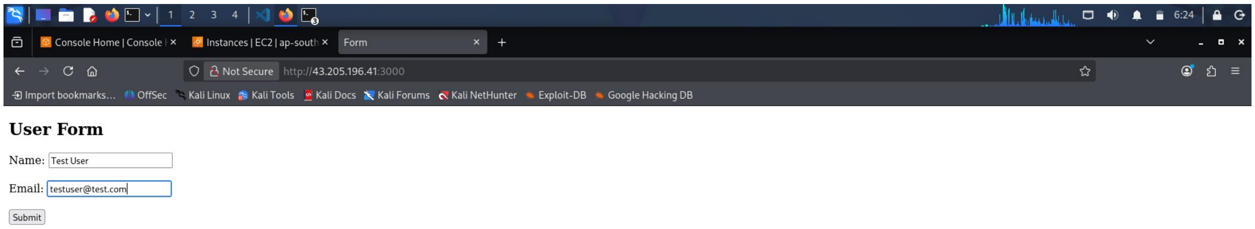
Part 2: Deploy Flask and Express on Separate EC2 Instances

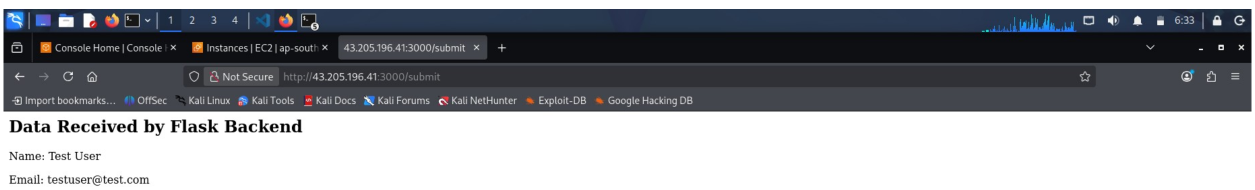
1. **Objective**: Deploy the Flask backend and the Express frontend on **two separate EC2 instances** using Terraform.
    - Provision two EC2 instances using Terraform:
        - One for the Flask backend.
        - One for the Express frontend.
    - Configure security groups to:
        - Allow communication between the two instances.
        - Expose both applications to the internet on their respective ports.
2. **Configuration**:
    - Use Terraform to define networking resources, such as VPC, subnets, and security groups.
    - Use user data scripts to automate the installation and startup of both applications.
3. **Expected Deliverables**:
    - Terraform configuration files.
    - Two working EC2 instances: one running Flask and one running Express.
    - Security groups configured to allow proper communication and public access.

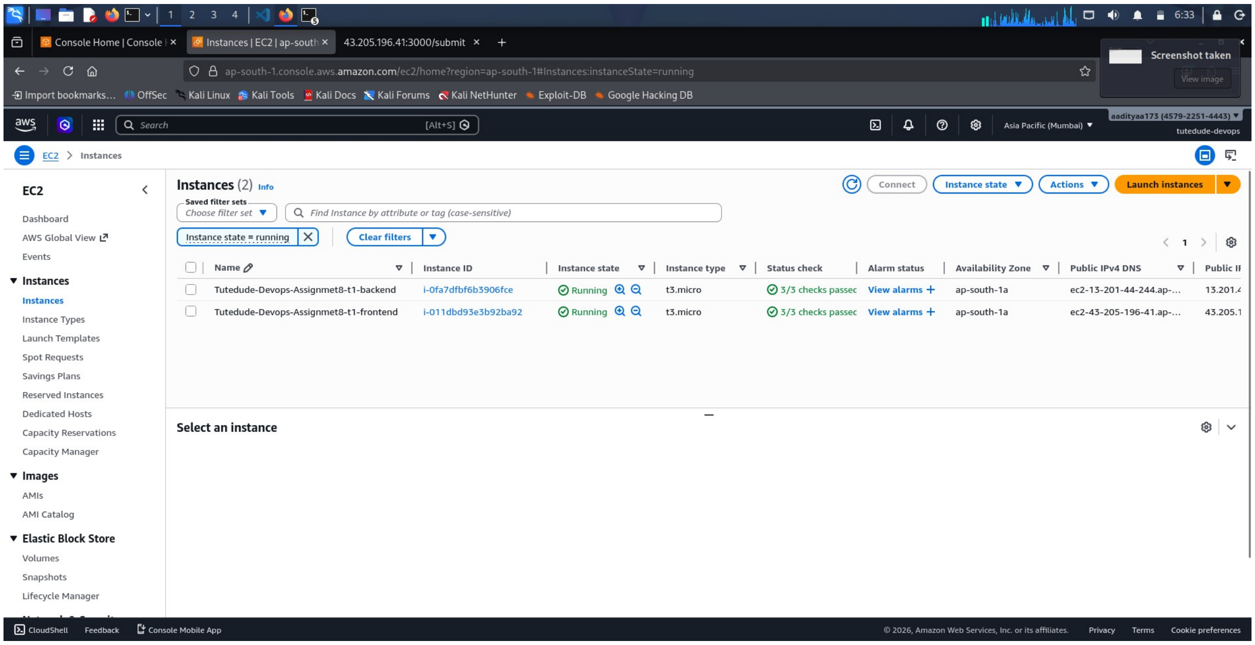



Home
File System
Trash
Tutedude
kali@kali: ~/Desktop/Tutedude/tutedude-devops-assignments/Assignment-8/task-2/terraform
Session Actions Edit View Help
+ tags_all = {
+ "Name" = "terraform-sg"
}
+ vpc_id = (known after apply)
}
Plan: 5 to add, 0 to change, 0 to destroy.
Changes to Outputs:
+ ec2_private_dns = [
+ (known after apply),
+ (known after apply),
]
+ ec2_private_ip = [
+ (known after apply),
+ (known after apply),
]
+ ec2_public_dns = [
+ (known after apply),
+ (known after apply),
]
+ ec2_public_ip = [
+ (known after apply),
+ (known after apply),
]
Note: You didn't use the -out option to save this plan, so Terraform
can't guarantee to take exactly these actions if you run "terraform
apply" now.
┌──(kali㉿kali)-[~/…/tutedude-devops-assignments/Assignment-8/task-2/terr
aform]
└─$


kali@kali: ~/Desktop/Tutedude/tutedude-devops-assignments/Assignment-8/task-2/ansible/playbooks/backend
Session Actions Edit View Help
┌──(kali㉿kali)-[~/…/task-2/ansible/playbooks/backend]
└─$ ansible-playbook -i inventry setup.yaml
PLAY [update and setup the ubuntu server] ********************************
****
TASK [Gathering Facts] **
****
[WARNING]: Host '13.201.44.244' is using the discovered Python interprete
r at '/usr/bin/python3.12', but future installation of another Python int
erpreter could cause a different interpreter to be discovered. See https:
//docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_dis
covery.html for more information.
ok: [13.201.44.244]
TASK [Update apt cache] ***
****
changed: [13.201.44.244]
TASK [Install the python] ***
****
changed: [13.201.44.244]
PLAY RECAP **
****
13.201.44.244 : ok=3 changed=2 unreachable=0 failed
=0 skipped=0 rescued=0 ignored=0
┌──(kali㉿kali)-[~/…/task-2/ansible/playbooks/backend]
└─$
┌──(kali㉿kali)-[~/…/task-2/ansible/playbooks/backend]
kali@kali: ~/Desktop/Tutedude/tutedude-devops-assignments/Assignment-8/task-2/terraform
Session Actions Edit View Help
aws_instance.tutedude-devops-assignment["Tutedude-Devops-Assignmet8-t1-fr
ontend"]: Still creating... [10s elapsed]
aws_instance.tutedude-devops-assignment["Tutedude-Devops-Assignmet8-t1-ba
ckend"]: Still creating... [20s elapsed]
aws_instance.tutedude-devops-assignment["Tutedude-Devops-Assignmet8-t1-fr
ontend"]: Still creating... [20s elapsed]
aws_instance.tutedude-devops-assignment["Tutedude-Devops-Assignmet8-t1-fr
ontend"]: Creation complete after 22s [id=i-011dbd93e3b92ba92]
aws_instance.tutedude-devops-assignment["Tutedude-Devops-Assignmet8-t1-ba
ckend"]: Creation complete after 23s [id=i-0fa7dfbf6b3906fce]
Apply complete! Resources: 5 added, 0 changed, 0 destroyed.
Outputs:
ec2_private_dns = [
"ip-172-31-46-217.ap-south-1.compute.internal",
"ip-172-31-43-102.ap-south-1.compute.internal",
]
ec2_private_ip = [
"172.31.46.217",
"172.31.43.102",
]
ec2_public_dns = [
"ec2-13-201-44-244.ap-south-1.compute.amazonaws.com",
"ec2-43-205-196-41.ap-south-1.compute.amazonaws.com",
]
ec2_public_ip = [
"13.201.44.244",
"43.205.196.41",
]
┌──(kali㉿kali)-[~/…/tutedude-devops-assignments/Assignment-8/task-2/terr
aform]
└─$


Console Home | Console
Instances | EC2 | ap-south
Form
Not Secure
http://43.205.196.41:3000
Import bookmarks...
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
6:24
User Form
Name: Test User
Email: testuser@test.com
Submit


Console Home | Console
Instances | EC2 | ap-south
43.205.196.41:3000/submit
Not Secure
http://43.205.196.41:3000/submit
Import bookmarks...
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
6:33
Data Received by Flask Backend
Name: Test User
Email: testuser@test.com


Console Home | Console
Instances | EC2 | ap-south
43.205.196.41:3000/submit
ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#Instances:instanceState=running
Screenshot taken
View image
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Search [Alt+S]
Asia Pacific (Mumbai)
tutedude-devops
EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Instances (2) Info
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Find Instance by attribute or tag (case-sensitive)
Instance state = running
Clear filters
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4 DNS
Tutedude-Devops-Assignmet8-t1-backend
i-0fa7dfbf6b3906fce
Running
t3.micro
3/3 checks passec
View alarms
ap-south-1a
ec2-13-201-44-244.ap-...
Tutedude-Devops-Assignmet8-t1-frontend
i-011dbd93e3b92ba92
Running
t3.micro
3/3 checks passec
View alarms
ap-south-1a
ec2-43-205-196-41.ap-...
Select an instance
CloudShell
Feedback
Console Mobile App
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Cookie preferences